# Earnshaw Balls — Solution

Y25 (2025) — English translation

**A1** **0.50 pts**

*Express $\ddot{\vec{r}}$ and $\dot{\vec{\omega}}$ in terms of $\vec{F}$, $m$, $\beta$, $\vec{a}$.*

Let's write down Newton's second law:

$$m\ddot{\vec{r}} = m\vec{g} + \vec{N} + \vec{F}.$$

Ball/Sphere не движется в вертикальном направлении, therefore $m\vec{g} + \vec{N} = 0$. We get:

The second answer follows from the law of change in angular momentum relative to the center of the ball:

**Answer:** \textbfAnswer:

$$\ddot{\vec{r}} = \frac{\vec{F}}{m}$$

**A2** **0.30 pts**

*Write down the condition for kinematic coupling. Express your answer in terms of $\dot{\vec{r}}$, $\vec{a}$, $\vec{r}$, $\vec{\omega}$ and $\vec{\Omega}$.*

The velocities of the table and the ball at the point of their contact must match, therefore:

**Answer:** \textbfAnswer:

$$\left[\vec{\Omega}, \vec{r}\right] = \dot{\vec{r}} + [\vec{\omega}, \vec{a}]$$

**A3** **0.70 pts**

*Express $\ddot{\vec{r}}$ in terms of $\beta$, $\dot{\vec{r}}$ and $\vec{\Omega}$.*

Let's express $\vec{F}$ from the equation by $\dot{\vec{\omega}}$, multiplying it vectorially by $\vec{a}$:

$$\beta m a^2 \left[\dot{\vec{\omega}}, \vec{a}\right] = \left[\left[\vec{a}, \vec{F}\right], \vec{a}\right] = \left[\vec{a}, \left[\vec{F}, \vec{a}\right]\right] = a^2 \vec{F}.$$

$$\vec{F} = \beta m \left[\dot{\vec{\omega}}, \vec{a}\right].$$

Let us express via/through заданные величины, продифференцировав кинематическую relation:

$$\left[\vec{\Omega}, \dot{\vec{r}}\right] = \ddot{\vec{r}} + \left[\dot{\vec{\omega}}, \vec{a}\right].$$

$$\vec{F} = \beta m \left[\dot{\vec{\omega}}, \vec{a}\right] = \beta m \left(\left[\vec{\Omega}, \dot{\vec{r}}\right] - \ddot{\vec{r}}\right).$$

Substitute the resulting expression and get the answer:

**Answer:** \textbfAnswer:

$$\ddot{\vec{r}} = \frac{\beta}{1+\beta} \left[\vec{\Omega}, \dot{\vec{r}}\right]$$

**A4** **0.70 pts**

*Let the initial position and speed of the ball be $\vec{r}_0$ and $\vec{v}_0$, respectively. Determine how $\dot{\vec{r}}$ depends on $\vec{r}$ as it moves further. Qualitatively depict the trajectory of the center of the ball in this case and indicate the characteristic dimensions and coordinates.*

Note that the expression obtained in the previous paragraph is the equality of two total derivatives with respect to time:

$$\frac{\mathrm{d}}{\mathrm{d}t}\dot{\vec{r}} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\beta}{1+\beta}\left[\vec{\Omega},\vec{r}\right].$$

Having integrated это coratio taking into account того, что velocity центра ball/spherea всегда vaporаллельна плоскости стола:

$$\dot{\vec{r}} = \frac{\beta}{1+\beta}\left[\vec{\Omega},\vec{r}-\vec{R}_0\right],$$

where $\vec{R}_0$ - постоянный vector. Note that это equation заgives движение по окружности с центром $\vec{R}_0$. Now let's substitute the initial conditions. Then integrating the acceleration will give us:

As we have already seen, the trajectory is a circle. All that remains is to find its center and radius.

$$\dot{\vec{r}} = \frac{\beta}{1+\beta}\left[\vec{\Omega},\vec{r}-\vec{R}_0\right] = \frac{\beta}{1+\beta}\left[\vec{\Omega},\vec{r}-\vec{r}_0\right]+\vec{v}_0$$

Домножим vectorно на $\vec{\Omega}$:

$$\frac{\beta}{1+\beta}\left[\vec{\Omega},\left[\vec{\Omega},\vec{r}-\vec{R}_0\right]\right] = \frac{\beta}{1+\beta}\left[\vec{\Omega},\left[\vec{\Omega},\vec{r}-\vec{r}_0\right]\right]+\left[\vec{\Omega},\vec{v}_0\right].$$

$$\frac{\beta}{1+\beta}\left[\vec{\Omega},\left[\vec{\Omega},\vec{r}_0-\vec{R}_0\right]\right] = \frac{\beta\Omega^2}{1+\beta}\left(\vec{R}_0-\vec{r}_0\right) = \left[\vec{\Omega},\vec{v}_0\right]$$

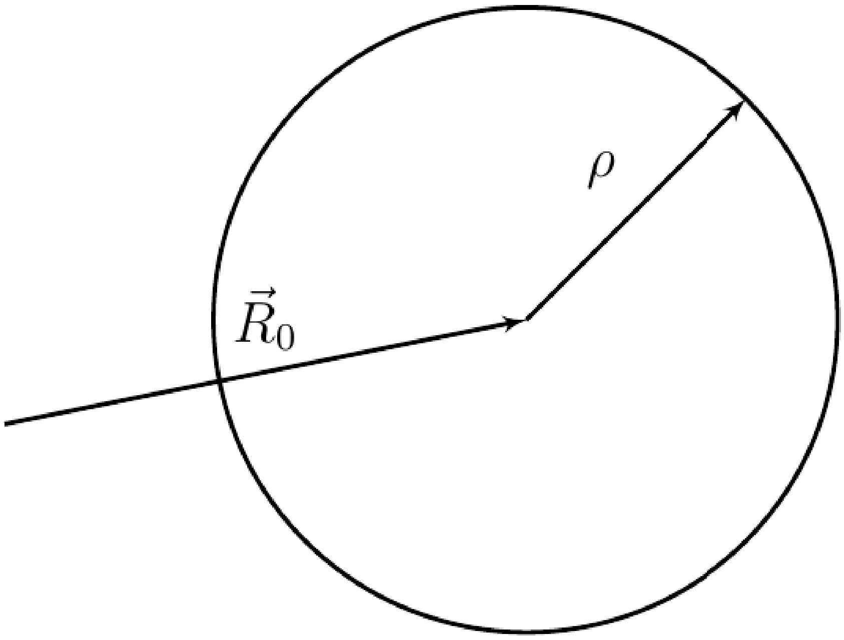
$$\vec{R}_0-\vec{r}_0 = \frac{1+\beta}{\Omega^2\beta}\left[\vec{\Omega},\vec{v}_0\right]$$

From погауенного соотношения ccy we get центр and radius окружности:

$$\vec{R}_0 = \vec{r}_0+\frac{1+\beta}{\Omega^2\beta}\left[\vec{\Omega},\vec{v}_0\right], \qquad \rho = \frac{(1+\beta)\,v_0}{\Omega\beta}.$$

**Answer:** \textbfAnswer:

$$\dot{\vec{r}} = \frac{\beta}{1+\beta}\left[\vec{\Omega},\vec{r}-\vec{r}_0\right]+\vec{v}_0$$




## B1 — 0.30 pts

*Express $\ddot{\vec{r}}$ in terms of $\vec{g}_{\|}$, $\beta$, $\dot{\vec{r}}$ and $\vec{\Omega}$.*

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The only difference from the previous part is that in the initial expression for $\ddot{\vec{r}}$ there will be a non-zero contribution $m\vec{g} + \vec{N} = m\vec{g}_{\|}$.

$$\ddot{\vec{r}} = \vec{g}_{\|} + \beta\left(\left[\vec{\Omega}, \dot{\vec{r}}\right] - \ddot{\vec{r}}\right)$$

**Answer:** \textbfAnswer:

$$\ddot{\vec{r}} = \frac{\vec{g}_{\|}}{1+\beta} + \frac{\beta}{1+\beta}\left[\vec{\Omega}, \dot{\vec{r}}\right]$$

## B2 — 1.40 pts

*Let us denote*

$$\vec{u} = \frac{\left[\vec{\Omega} \times \vec{g}_{\|}\right]}{\Omega^2}.$$

*Качественно изобtimesandme траектории центра ball/spherea for/at следующих начальных скоростях: $\vec{v}_0 = \vec{u}$ $\vec{v}_0 = 2.5\vec{u}$ $\vec{v}_0 = 5\vec{u}$ $\vec{v}_0 = 0$ For the case 4 рассчитайте численно характерные смеры траектории for $g_{\|} = 1.0\ m/s^2$, $\Omega = 10\ rad/s$.*

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For case 4, numerically calculate the characteristic dimensions of the trajectory for $g_{\|} = 1.0\ m/s^2$, $\Omega = 10$ rad/.

Let's analyze the answer from the previous paragraph. Acceleration consists of a constant term and a term obtained by the vector product of speed and a constant vector. This is completely analogous to movement in a crossed magnetic and electric field. Let's find the drift speed $\dot{\vec{r}} = \vec{v} + \vec{v}$.

$$\dot{\vec{v}} = \frac{\vec{g}_{\|}}{1+\beta} + \frac{\beta}{1+\beta}\left[\vec{\Omega}, \vec{v}\right] + \frac{\beta}{1+\beta}\left[\vec{\Omega}, \vec{v}\right]$$

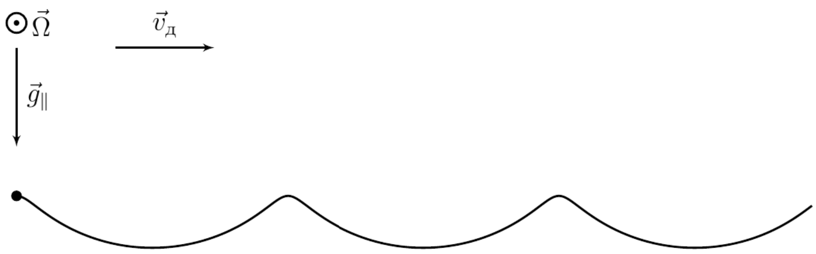
$$0 = \frac{\vec{g}_{\|}}{1+\beta} + \frac{\beta}{1+\beta}\left[\vec{\Omega}, \vec{v}\right]$$

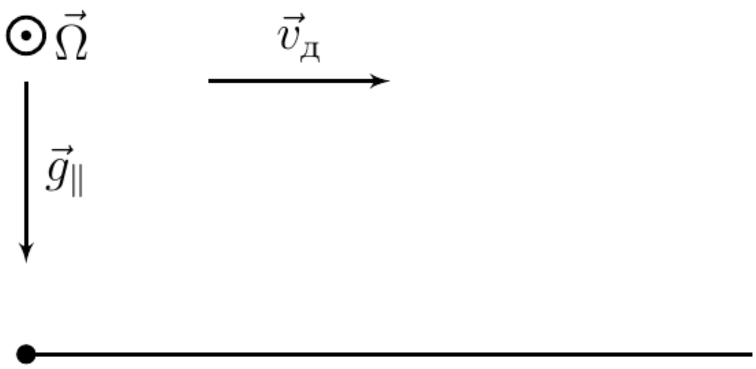
Vectorно домножив на $\vec{\Omega}$, we get value скорости дрейфа:

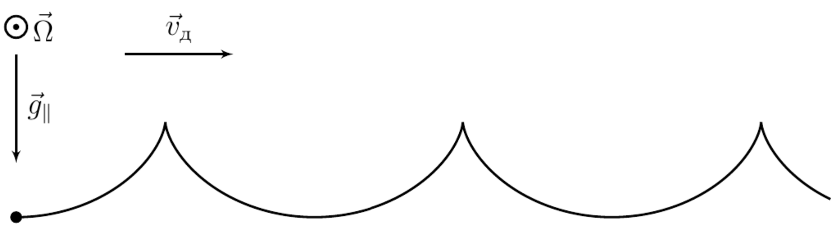
$$\vec{v} = \frac{\left[\vec{\Omega}, \vec{g}_{||}\right]}{\Omega^2 \beta} = \frac{5}{2}\vec{u}.$$

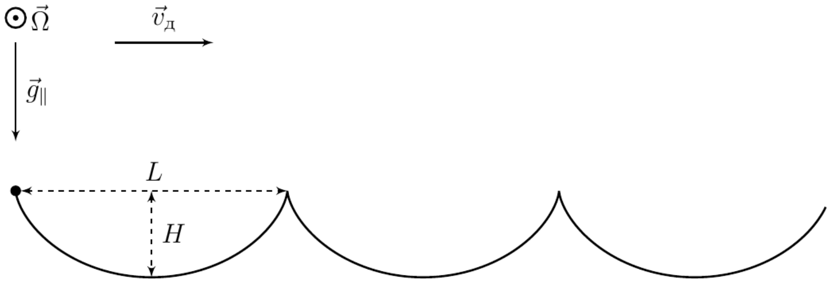
Vector $\vec{v}$ вращается с постоянной угловой velocityую $\beta\vec{\Omega}/(1+\beta)$. Using the obtained expressions, we obtain the form of the trajectory for each case.

**Answer:** \textbfAnswer: Case 1: the difference from the drift speed is small, so there will be a trajectory without self-intersections.



## C1 0.20 pts

*Express $\ddot{\vec{r}}$ in terms of $m$, $\beta$, $\vec{a}$, $\vec{r}$, $\vec{N}$, $\vec{g}$, $\dot{\vec{\omega}}$*

From Newton's second law:

$$\ddot{\vec{r}} = \frac{\vec{N}}{m} + \vec{g} + \frac{\vec{F}}{m}.$$

From the law изменения momenta momentuma:

$$ma^2\beta\dot{\vec{\omega}} = \left[\vec{a} \times \vec{F}\right] \quad \Rightarrow \quad ma^2\beta\left[\vec{a} \times \dot{\vec{\omega}}\right] = \left[\vec{a} \times \left[\vec{a} \times \vec{F}\right]\right] = \vec{a}\left(\vec{a} \cdot \vec{F}\right) - \vec{F}a^2 = -\vec{F}a^2.$$

From:

**Answer:** \textbfAnswer:

$$\ddot{\vec{r}} = \frac{\vec{N}}{m} + \vec{g} - \left[\vec{a} \times \dot{\vec{\omega}}\right]\beta$$

## C2 0.40 pts

*Write down the condition for kinematic coupling. Express your answer in terms of $\dot{\vec{r}}$, $\vec{a}$, $\vec{r}$, $\vec{\omega}$, $\vec{\Omega}$.*

The velocities of the table and the ball at the point of their contact must coincide, therefore:

$$\dot{\vec{r}} + [\vec{\omega} \times \vec{a}] = \left[\vec{\Omega} \times (\vec{r} + \vec{a})\right],$$

from:

**Answer:** \textbfAnswer:

$$\dot{\vec{r}} = \left[\vec{\Omega} \times (\vec{r} + \vec{a})\right] - [\vec{\omega} \times \vec{a}]$$

## C3 0.90 pts

*Express $\ddot{\vec{r}}$ in terms of $m$, $\beta$, $\vec{a}$, $\vec{N}$, $\vec{g}$, $\dot{\vec{r}}$, $\vec{\Omega}$, $\vec{\zeta}$, $\vec{\omega}$.*

Let us differentiate the expression for $\dot{\vec{r}}$ with respect to time:

$$\ddot{\vec{r}} = \left[\vec{\Omega} \times \dot{\vec{r}}\right] + \left[\vec{\Omega} \times \dot{\vec{a}}\right] - [\dot{\vec{\omega}} \times \vec{a}] - [\vec{\omega} \times \dot{\vec{a}}]$$

From выражения for $\ddot{\vec{r}}$, погауенного в parte C1 :

$$[\dot{\vec{\omega}} \times \vec{a}] = \frac{1}{\beta}\left(\ddot{\vec{r}} - \frac{\vec{N}}{m} - \vec{g}\right)$$

Also:

$$\dot{\vec{a}} = \left[\vec{\zeta} \times \vec{a}\right],$$

from where:

Let's see how this formula is transformed taking into account some approximations, namely: $\theta \ll 1$, that is, the table is slightly different from a flat one and the influence of gravity is weak; the motion of the ball is finite, that is, it moves in a limited region of space near the vertex and does not move away to infinity; $a \lesssim r$, that is, the ball is not too close to the top of the cone; $\zeta \lesssim \Omega$, that is, the rotation of the table occurs quite quickly in comparison with the speed of the ball; $|(\vec{\omega}, \vec{a})| \lesssim \Omega r$, that is, the spinning speed is not too high. Let us restrict ourselves to terms of order of smallness $\theta$. Note that the terms with $\vec{N}$ and $\vec{g}$ have the order of smallness $\theta$, since the projection $\vec{g}$ onto the surface of the cone is of this order. Due to the fact that the motion is finite, the remaining terms must be of the same order. Now it remains to show that the first term is an order of magnitude larger than the last, then the first is of the first order of smallness, and the last is of the second, and it can be discarded. Estimation of the value of the first term:

$$\left|\beta\frac{[\vec{\Omega},\dot{\vec{r}}]}{1+\beta}\right| \sim \left|[\vec{\Omega},\dot{\vec{r}}]\right| \sim \Omega\dot{r}+\Omega\zeta r.$$

Теперь последнее term expanding на два слагаемых and проаналfromandруем каждое from них taking into account того, что angle между $\vec{\zeta}$ and $\vec{a}$ equals $\theta$:

$$\left|\frac{\beta}{1+\beta}\left[\vec{\Omega},\left[\vec{\zeta},\vec{a}\right]\right]\right| \sim \Omega\zeta a\theta \lesssim \Omega\zeta r\theta \ll \Omega\zeta r \lesssim \left|\beta\frac{[\vec{\Omega},\dot{\vec{r}}]}{1+\beta}\right|.$$

For оценки второго слагаемого expanding $\vec{\omega}$ на компоненты вдоль and перпендикулярно $\vec{a}$: $\vec{\omega}=\vec{\omega}_a+\vec{\omega}_\perp$. Note that from кинетической связи and последнего условия for/atближения:

$$\omega a \lesssim \left|\dot{\vec{r}}\right| + \Omega r + |(\vec{\omega},\vec{a})| \sim \dot{r}+\Omega r.$$

Thus последнее expression, малость которого надо доказать:

$$\left|\frac{\beta}{1+\beta}\left[\vec{\omega},\left[\vec{\zeta},\vec{a}\right]\right]\right| \sim \omega\zeta a\theta \lesssim \left(\dot{r}+\Omega r\right)\zeta\theta \ll \Omega\dot{r}+\Omega\zeta r \sim \left|\beta\frac{[\vec{\Omega},\dot{\vec{r}}]}{1+\beta}\right|.$$

Thus, we obtain the desired approximate formula.

Let us restrict ourselves to terms of order of smallness $\theta$.

Note that the terms with $\vec{N}$ and $\vec{g}$ have the order of smallness $\theta$, since the projection $\vec{g}$ onto the surface of the cone is of this order. Due to the fact that the motion is finite, the remaining terms must be of the same order. Now it remains to show that the first term is an order of magnitude larger than the last, then the first is of the first order of smallness, and the last is of the second, and it can be discarded.

Estimation of the value of the first term:

$$\left|\beta\frac{[\vec{\Omega},\dot{\vec{r}}]}{1+\beta}\right| \sim \left|[\vec{\Omega},\dot{\vec{r}}]\right| \sim \Omega\dot{r}+\Omega\zeta r.$$

Now let's decompose the last term into two terms and analyze each of them, taking into account the fact that the angle between $\vec{\zeta}$ and $\vec{a}$ is equal to $\theta$:

$$\left|\frac{\beta}{1+\beta}\left[\vec{\Omega},\left[\vec{\zeta},\vec{a}\right]\right]\right| \sim \Omega\zeta a\theta \lesssim \Omega\zeta r\theta \ll \Omega\zeta r \lesssim \left|\beta\frac{[\vec{\Omega},\dot{\vec{r}}]}{1+\beta}\right|.$$

To estimate the second term, we decompose $\vec{\omega}$ into components along and perpendicular to $\vec{a}$: $\vec{\omega}=\vec{\omega}_a+\vec{\omega}_\perp$. Note that from the kinetic connection and the last approximation condition:

$$\omega a \lesssim \left|\dot{\vec{r}}\right| + \Omega r + |(\vec{\omega},\vec{a})| \sim \dot{r}+\Omega r.$$

Thus, the last expression, the smallness of which must be proven:

$$\left|\frac{\beta}{1+\beta}\left[\vec{\omega},\left[\vec{\zeta},\vec{a}\right]\right]\right| \sim \omega\zeta a\theta \lesssim \left(\dot{r}+\Omega r\right)\zeta\theta \ll \Omega\dot{r}+\Omega\zeta r \sim \left|\beta\frac{[\vec{\Omega},\dot{\vec{r}}]}{1+\beta}\right|.$$

Thus we obtain the desired approximate formula.

**Answer:** \textbfAnswer:

$$\ddot{\vec{r}} = \frac{[\vec{\Omega}\times\dot{\vec{r}}]}{1+\beta^{-1}} + \frac{1}{1+\beta}\left(\frac{\vec{N}}{m}+\vec{g}\right) + \frac{1}{1+\beta^{-1}}\left[\left(\vec{\Omega}-\vec{\omega}\right)\times\left[\vec{\zeta}\times\vec{a}\right]\right]$$

## C4 **1.00 pts**

*Prove that in the considered approximation the quantity*

$$l = r^2\left(\zeta - \frac{\Omega\beta}{2\left(1+\beta\right)}\right).$$

*Here $\zeta = |\vec{\zeta}|$ is conserved.*

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First decision. Let us expand the terms from this equation according to the introduced basis:

$$\dot{\vec{r}} = \left[\vec{\zeta}\times\vec{r}\right] + \dot{r}\vec{e}_r = \zeta r\vec{e}_\varphi + \dot{r}\vec{e}_r$$

$$\ddot{\vec{r}} = \left(\dot{\zeta}r\vec{e}_\varphi + \zeta\dot{r}\vec{e}_\varphi + \zeta r\frac{\mathrm{d}\vec{e}_\varphi}{\mathrm{d}t}\right) + \left(\ddot{r}\vec{e}_r + \dot{r}\frac{\mathrm{d}\vec{e}_r}{\mathrm{d}t}\right) = = \dot{\zeta}r\vec{e}_\varphi + \zeta\dot{r}\vec{e}_\varphi + \zeta^2 r\left(-\vec{e}_r - \theta\vec{e}_z\right) + \ddot{r}\vec{e}_r + \zeta\dot{r}\vec{e}_\varphi = \left(\dot{\zeta}r + 2\zeta\dot{r}\right)\vec{e}_\varphi + \left(\ddot{r} - \zeta^2 r\right)$$

$$\left[\vec{\Omega}\times\dot{\vec{r}}\right] = \Omega\dot{r}\vec{e}_\varphi - \Omega\zeta r\vec{e}_r + \Omega\zeta r\theta\vec{e}_z$$

$$\frac{\vec{N}}{m} + \vec{g} = g\theta\vec{e}_r + \left(\frac{N_z}{m} - g\right)\vec{e}_z$$

Теперь substituting их в данное equation, взяв проекции на $\vec{e}_\varphi$:

$$\dot{\zeta}r + 2\zeta\dot{r} = \frac{\Omega\dot{r}}{1+\beta^{-1}} \quad\Rightarrow\quad \frac{\mathrm{d}\zeta}{\mathrm{d}r} = \frac{1}{r}\left(\frac{\Omega}{1+\beta^{-1}} - 2\zeta\right) \quad\Rightarrow\quad \int\frac{\mathrm{d}\zeta}{\dfrac{\Omega}{1+\beta^{-1}} - 2\zeta} = \int\frac{\mathrm{d}r}{r} \quad\Rightarrow\quad r^2\left(\zeta - \frac{\Omega\beta}{2\left(1+\beta\right)}\right) = \mathrm{const}$$

which is what needed to be proved.


Let us expand the terms from this equation according to the introduced basis:

$$\dot{\vec{r}} = \left[\vec{\zeta}\times\vec{r}\right] + \dot{r}\vec{e}_r = \zeta r\vec{e}_\varphi + \dot{r}\vec{e}_r$$

$$\ddot{\vec{r}} = \left(\dot{\zeta}r\vec{e}_\varphi + \zeta\dot{r}\vec{e}_\varphi + \zeta r\frac{\mathrm{d}\vec{e}_\varphi}{\mathrm{d}t}\right) + \left(\ddot{r}\vec{e}_r + \dot{r}\frac{\mathrm{d}\vec{e}_r}{\mathrm{d}t}\right) = = \dot{\zeta}r\vec{e}_\varphi + \zeta\dot{r}\vec{e}_\varphi + \zeta^2 r\left(-\vec{e}_r - \theta\vec{e}_z\right) + \ddot{r}\vec{e}_r + \zeta\dot{r}\vec{e}_\varphi = \left(\dot{\zeta}r + 2\zeta\dot{r}\right)\vec{e}_\varphi + \left(\ddot{r} - \zeta^2 r\right)$$

$$\left[\vec{\Omega}\times\dot{\vec{r}}\right] = \Omega\dot{r}\vec{e}_\varphi - \Omega\zeta r\vec{e}_r + \Omega\zeta r\theta\vec{e}_z$$

$$\frac{\vec{N}}{m} + \vec{g} = g\theta\vec{e}_r + \left(\frac{N_z}{m} - g\right)\vec{e}_z$$

Теперь substituting их в данное equation, взяв проекции на $\vec{e}_\varphi$:

$$\dot{\zeta}r + 2\zeta\dot{r} = \frac{\Omega\dot{r}}{1+\beta^{-1}} \quad\Rightarrow\quad \frac{\mathrm{d}\zeta}{\mathrm{d}r} = \frac{1}{r}\left(\frac{\Omega}{1+\beta^{-1}} - 2\zeta\right) \quad\Rightarrow\quad \int\frac{\mathrm{d}\zeta}{\dfrac{\Omega}{1+\beta^{-1}} - 2\zeta} = \int\frac{\mathrm{d}r}{r} \quad\Rightarrow\quad r^2\left(\zeta - \frac{\Omega\beta}{2\left(1+\beta\right)}\right) = \mathrm{const}$$

which is what needed to be proved.


Second solution. Let's consider the change in the angular momentum of the center of mass:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\vec{r},\dot{\vec{r}}\right] = \left[\vec{r},\ddot{\vec{r}}\right].$$

For проекции на ось $z$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\vec{e}_z,\vec{r},\dot{\vec{r}}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\zeta\right) = \left(\vec{e}_z,\vec{r},\ddot{\vec{r}}\right).$$

Note that $\left(\vec{e}_z,\vec{r},\vec{N}\right)$ and $\left(\vec{e}_z,\vec{r},\vec{g}\right)$ equal нулю, since эти тройки vectorов лежат в одной плоскости, therefore от ускорения останется только одно term:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\zeta\right) = \frac{\beta}{1+\beta}\left(\vec{e}_z,\vec{r},\ [\vec{\Omega},\dot{\vec{r}}]\right) = \frac{\beta}{1+\beta}\left(\vec{e}_z,\vec{\Omega}\left(\vec{r},\dot{\vec{r}}\right) - \dot{\vec{r}}\left(\vec{\Omega},\vec{r}\right)\right).$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\zeta\right) = \frac{\beta}{1+\beta}\left(\left(\vec{e}_z, \vec{\Omega}\right)\left(\vec{r}, \dot{\vec{r}}\right) - \left(\vec{e}_z, \dot{\vec{r}}\right)\left(\vec{\Omega}, \vec{r}\right)\right)$$

Substituting значения скальных произведений and отбросим terms второго порядка малости:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\zeta\right) = \frac{\beta}{1+\beta}\left(\Omega\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{r^2}{2}\right) - r\dot{r}\Omega\theta^2\right) \approx \frac{\beta\Omega}{1+\beta}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{r^2}{2}\right).$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\left(\zeta - \frac{\Omega\beta}{2\left(1+\beta\right)}\right)\right) = 0$$

Поrayandли, что quantity $l$ is saved.

Let's consider the change in the angular momentum of the center of mass:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\vec{r}, \dot{\vec{r}}\right] = \left[\vec{r}, \ddot{\vec{r}}\right].$$

For projection onto the $z$ axis:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\vec{e}_z, \vec{r}, \dot{\vec{r}}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\zeta\right) = \left(\vec{e}_z, \vec{r}, \ddot{\vec{r}}\right).$$

Note that $\left(\vec{e}_z, \vec{r}, \vec{N}\right)$ and $\left(\vec{e}_z, \vec{r}, \vec{g}\right)$ are equal to zero, since these triples of vectors lie in the same plane, so only one term will remain from the acceleration:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\zeta\right) = \frac{\beta}{1+\beta}\left(\vec{e}_z, \vec{r},\ [\vec{\Omega}, \dot{\vec{r}}]\right) = \frac{\beta}{1+\beta}\left(\vec{e}_z, \vec{\Omega}\left(\vec{r}, \dot{\vec{r}}\right) - \dot{\vec{r}}\left(\vec{\Omega}, \vec{r}\right)\right).$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\zeta\right) = \frac{\beta}{1+\beta}\left(\left(\vec{e}_z, \vec{\Omega}\right)\left(\vec{r}, \dot{\vec{r}}\right) - \left(\vec{e}_z, \dot{\vec{r}}\right)\left(\vec{\Omega}, \vec{r}\right)\right)$$

Let us substitute the values of the rock products and discard the terms of the second order of smallness:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\zeta\right) = \frac{\beta}{1+\beta}\left(\Omega\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{r^2}{2}\right) - r\dot{r}\Omega\theta^2\right) \approx \frac{\beta\Omega}{1+\beta}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{r^2}{2}\right).$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\left(\zeta - \frac{\Omega\beta}{2\left(1+\beta\right)}\right)\right) = 0$$

We found that the value $l$ is preserved.

**C5** **0.50 pts**

*Express $\ddot{r}$ in terms of $r$, $\zeta$, $\beta$, $\Omega$, $\theta$ and $g$.*

Now consider the equation:

$$\ddot{\vec{r}} = \frac{1}{1+\beta}\left(\vec{g} + \frac{\vec{N}}{m}\right) + \frac{\beta\left[\vec{\Omega}\times\dot{\vec{r}}\right]}{1+\beta}.$$

Substituting в него проекции слагаемых на $\vec{e}_r$:

$$\ddot{r} - \zeta^2 r = \frac{g\theta}{1+\beta} - \frac{\beta\Omega\zeta r}{1+\beta} \quad \Rightarrow \quad \ddot{r} = \zeta^2 r + \frac{g\theta}{1+\beta} - \frac{\beta\Omega\zeta r}{1+\beta}.$$

**Answer:** \textbfAnswer:

$$\ddot{r} = \zeta^2 r + \frac{g\theta}{1+\beta} - \frac{\beta\Omega\zeta r}{1+\beta}$$

**C6** **0.70 pts**

*Determine at what values of $r$ movement along circular trajectories around the axis of symmetry of the cone $(r = \text{const})$ is possible and determine the values of $\zeta$ corresponding to such movement. Express your answer in terms of $r$, $\beta$, $\Omega$, $\theta$ and $g$.*

$$r = \text{const} \quad \Rightarrow \quad \ddot{r} = 0 = \zeta^2 r + \frac{g\theta}{1+\beta} - \frac{\beta\Omega\zeta r}{1+\beta}.$$

Then we get квадратное equation отноительно $\zeta$:

$$\zeta^2 - \zeta \cdot \frac{\Omega}{1+\beta^{-1}} + \frac{g\theta}{(1+\beta)r} = 0 \quad \Rightarrow \quad \zeta_{1,2} = \frac{\Omega}{2(1+\beta^{-1})} \pm \sqrt{\left(\frac{\Omega}{2(1+\beta^{-1})}\right)^2 - \frac{g\theta}{(1+\beta)r}}.$$

Since подкоренное expression не отрицательно:

$$r \geq \frac{4g\theta(1+\beta)}{\Omega^2\beta^2} = r_{\min}.$$

**Answer:** \textbfAnswer:

$$\zeta_{1,2} = \frac{\Omega}{2(1+\beta^{-1})} \pm \sqrt{\left(\frac{\Omega}{2(1+\beta^{-1})}\right)^2 - \frac{g\theta}{(1+\beta)r}}$$

$$r \geq \frac{4g\theta(1+\beta)}{\Omega^2\beta^2}$$

**C7** **0.80 pts**

*Express $\dot{r}^2$ up to an arbitrary constant in terms of $l$, $r$, $g$, $\beta$, $\Omega$ and $\theta$. Qualitatively depict the appearance of phase diagrams $\dot{r}(r)$.*

It is easy to see that:

$$\ddot{r} = \frac{\mathrm{d}\dot{r}}{\mathrm{d}t} = \frac{\mathrm{d}\dot{r}}{\mathrm{d}r/\dot{r}} = \frac{\mathrm{d}(\dot{r}^2)}{2\mathrm{d}r} \quad \Rightarrow \quad \dot{r}^2 = 2\int \ddot{r}(r)\mathrm{d}r.$$

Мы знаем, что сохраняется quantity $l$, consequently:

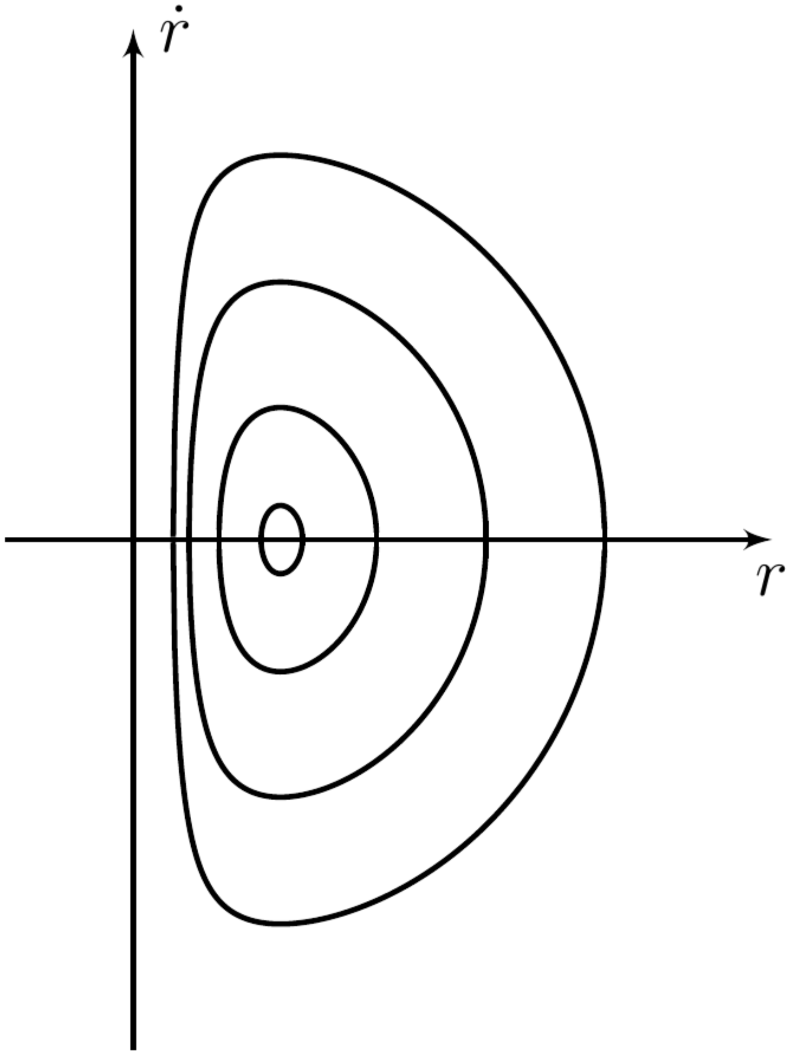
$$\zeta(r) = \frac{\Omega\beta}{2\,(1+\beta)} + \frac{l}{r^2}.$$

Substituting $\zeta$ в expression for $\ddot{r}$, we get:

$$\dot{r}^2 = 2\int\left(\frac{g\theta}{1+\beta} + \frac{l^2}{r^3} - \frac{\Omega^2 r}{4(1+\beta^{-1})^2}\right)\mathrm{d}r = \frac{2g\theta r}{1+\beta} - \frac{\Omega^2 r^2}{4(1+\beta^{-1})^2} - \frac{l^2}{r^2} + \text{const}.$$

**Answer:** \textbfAnswer:

$$\dot{r}^2 = \frac{2g\theta r}{1+\beta} - \frac{\Omega^2 r^2}{4(1+\beta^{-1})^2} - \frac{l^2}{r^2} + \text{const}$$




**C8** **0.40 pts**

*Let us consider a small perturbation of a circular orbit with radius $r_0$. Determine the cyclic frequency $\xi$ of small radial vibrations. Express your answer in terms of $\Omega$, $r_0$, $\theta$, $g$, $\beta$.*

The orbit was disturbed so that the distance from the axis to the center of the ball was presented in the form:

$$r(t) = r_0 + \delta(t), \ (\text{где } \delta \ll r_0 \text{ и } \langle r \rangle = r_0).$$

Then:

$$\ddot{\delta} = -\xi^2\delta = \ddot{r}(r_0) = \frac{2g\theta}{1+\beta} + \frac{l^2}{r^3} - \frac{\Omega^2 r}{4(1+\beta^{-1})^2} \approx -\frac{3l^2\delta}{r_0^4} - \frac{\Omega^2\delta}{4(1+\beta^{-1})^2},$$

hence:

$$\xi = \sqrt{\frac{3l^2}{r_0^4} + \frac{\Omega^2}{4(1+\beta^{-1})^2}},$$

where:

$$l = r_0^2\left(\zeta(r_0) - \frac{\Omega}{2(1+\beta^{-1})}\right) = \pm r_0^2\sqrt{\frac{\Omega^2}{4(1+\beta^{-1})^2} - \frac{g\theta}{(1+\beta)r_0}}.$$

Substituting $l$ в expression for $\xi$, we get: